
Curvature errors are low rank: aligned probing for Hessian-supervised interatomic potentials

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Abstract

Supervising a machine-learning interatomic potential on second derivatives is usually made affordable by replacing the $(3N)^2$ -entry Hessian loss with a Hutchinson estimate built from Hessian–vector products, one isotropic probe per configuration per minibatch. Isotropic probing is the variance-optimal choice only if the quantity being probed is isotropic. We show that it is not: for a trained neural-network potential on reactive C/H/O/N chemistry, the Hessian *error* $\Delta = \tilde{H} - H$ has a mean stable rank of 3.8 out of $3N \approx 42$, so a random probe spends most of its variance budget on directions in which the model is already correct. We turn this into an estimator. Each probe returns, for free, the residual vector Δv ; the last two dozen such vectors form a randomised Nyström sketch of Δ from which we extract a near-invariant subspace, and we draw the next probe from a two-stratum design aligned with it. The design is exactly unbiased for *any* subspace and any stratum probabilities, so a stale or wrong sketch costs efficiency but never correctness. At an unchanged probe count, aligned probing lowers the variance of the curvature-loss gradient to 0.55, 0.49 and 0.37 of the isotropic baseline on held-out reaction, intrinsic-reaction-coordinate and normal-mode-sampled geometries; the reduction is largest exactly where the model is worst. Spending that surplus on probes instead of on precision gives a $2.4\times$ reduction in probes per optimiser step at a gradient variance of 1.10–1.23 times the baseline’s and accuracy statistically indistinguishable from it on all nine (dataset $\times$ property) cells. Downstream accuracy in fact survives to $4.5\times$ fewer probes, past the point where per-step variance parity is lost, and a two-constant variance model predicts the ceiling we hit and identifies what sets it. Allocating probes unequally *across* molecules turns out to matter far less than allocating them across directions, and to help at all only once the Horvitz–Thompson weights it introduces are explicitly bounded—a conclusion our own early measurements got backwards, and which we document.

1 Introduction

Interatomic potentials learned from quantum-chemical reference data are trained almost universally on energies and forces. Second derivatives carry information that first derivatives do not: vibrational frequencies, zero-point energies, the character of a saddle point, and the curvature of a reaction path are all properties of the Hessian $H_{(i\alpha)(j\beta)} = \partial^2 E / \partial r_i^\alpha \partial r_j^\beta$, and models with excellent force accuracy can still misrepresent them [Loew et al., 2025]. The obstacle is cost. A Hessian loss has $(3N)^2$ terms per configuration, and differentiating it during training means backpropagating through all of them.

The now-standard remedy is to estimate the Hessian loss stochastically. Writing $\Delta = \tilde{H}_\theta - H$ for the error of the model Hessian and $\mathcal{L}_H = \|\Delta\|_F^2 / (3N)^2$, Hutchinson’s identity [Hutchinson, 1990] gives the unbiased estimate $\hat{\mathcal{L}}_H = \|\Delta v\|^2 / (3N)^2$ for any probe v with $\mathbb{E}[vv^\top] = I$, and Δv costs one Hessian–vector product [Pearlmutter, 1994], i.e. a small constant number of force evaluations. This is the idea behind projected Hessian learning [Rodriguez et al., 2026], and behind

related curvature-distillation and phonon-fine-tuning schemes [Amin et al., 2025, Koker et al., 2026, Cui et al., 2026]. In the standard protocol one probe is drawn per molecule per minibatch, so a minibatch of B structures costs B Hessian–vector products per optimiser step.

That protocol fixes an operating point on a variance–cost trade-off without asking whether it is the right one. The relevant question is not how many probes are affordable but how much information each probe buys, and the answer depends entirely on the geometry of the object being probed. An isotropic probe distributes its sampling effort uniformly over the $3N$ coordinate directions. That is optimal when the error Δ is itself isotropic, and increasingly wasteful as Δ concentrates.

The first contribution of this paper is the observation that Δ concentrates severely. For an ANI-style potential trained on reactive C/H/O/N chemistry, the mean stable rank $\|\Delta\|_F^2/\|\Delta\|_2^2$ of the Hessian error is 3.8 while $3N \approx 42$: three quarters of the squared error lives in a quarter of the directions. The model is not uniformly wrong; it is wrong in a few, molecule-specific curvature directions. Physically this is unsurprising—an atom-centred potential with a smooth radial basis reproduces soft collective motions far better than the stiff, strongly anharmonic modes of a bond being made or broken—but it has a sharp statistical consequence, which is our second contribution: the variance of the *gradient* of the Hutchinson curvature loss is a quadratic form in the tensor $S = \Delta^\top \partial_\theta \tilde{H}$, and a probe design that partitions the coordinate space and samples one part at a time annihilates all of the cross-part contributions to that quadratic form while paying only for the between-part heterogeneity of the mean. When the partition is aligned with the dominant error directions, the trade is strongly favourable.

The third contribution makes the design implementable. Aligning the probe requires knowing where the error lives, and computing that directly would defeat the purpose. It is not necessary: every probe already returns the residual vector $\rho = \Delta v$ as a by-product of the loss it computes, and $\mathbb{E}[\rho \rho^\top] = \Delta^2$. The last K pairs (v, ρ) of a molecule therefore constitute a randomised sketch of Δ , and its Nyström factorisation $\Delta \approx R(V^\top R)^+ R^\top$ yields a subspace that is *near-invariant* for Δ —the property that actually matters, and one that the cruder sketch $\sum_t \rho_t \rho_t^\top$ does not deliver. The estimator is exactly unbiased for any subspace and any stratum probabilities, so staleness and sketch error cost efficiency and never correctness; this is what makes an adaptive, self-supervised probe design safe to use inside an unbiased stochastic gradient method.

Empirically, aligned probing reduces the variance of the curvature-loss gradient by a factor of 1.8 to 2.7 at an unchanged probe count, with the largest gain on the far-from-equilibrium geometries where the curvature error is most concentrated. Spending that surplus on probes rather than on precision, we train with $2.4\times$ fewer Hessian–vector products per optimiser step, at a gradient variance of 1.10–1.23 times the baseline’s and accuracy indistinguishable from it on every energy, force and Hessian cell of three held-out evaluation sets. We also report where the idea stops: downstream accuracy survives to $4.5\times$ but per-step variance parity does not, a dissociation we quantify, and a two-constant variance model predicts the ceiling and shows that beyond it the binding term is no longer probe noise at all. Allocating probes unequally across *molecules*, which is where a reader of the stochastic-optimisation literature would look first, turns out to be worth about $1.3\times$ —real, but small next to the direction-space effect, and negative unless the inverse-probability weights it introduces are explicitly bounded.

2 The variance geometry of stochastic curvature supervision

Setup. A configuration of N atoms has coordinates $\mathbf{R} \in \mathbb{R}^{3N}$; write $n = 3N$. A potential $\tilde{E}_\theta$ predicts energies, forces $-\nabla_{\mathbf{R}} \tilde{E}_\theta$ and, implicitly, the Hessian $\tilde{H}_\theta = \nabla_{\mathbf{R}}^2 \tilde{E}_\theta$. With reference Hessian H and error $\Delta = \tilde{H}_\theta - H$, the per-configuration curvature loss and its Hutchinson estimate are

$$\ell(\theta) = \frac{\|\Delta\|_F^2}{n^2}, \quad \hat{\ell}(\theta; v) = \frac{\|\Delta v\|^2}{n^2}, \quad \mathbb{E}[v v^\top] = I_n, \quad (1)$$

and $\mathbb{E}_v[\hat{\ell}] = \text{tr}(\Delta^\top \Delta)/n^2 = \ell$. Only $\Delta v = \tilde{H}_\theta v - H v$ is needed, and $\tilde{H}_\theta v$ is one reverse-over-reverse Hessian–vector product. A minibatch of B structures contributes $\mathcal{L}_H = B^{-1} \sum_m \ell_m$ to the training objective $\mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \mathcal{L}_H$.

The gradient variance is a quadratic form in $S = \Delta^\top J$. Let $J_{ik} = \partial \tilde{H}_{ik} / \partial \theta$ and define the third-order tensor $S_{jk} = \sum_i \Delta_{ij} J_{ik}$, whose entries are vectors in parameter space. Differentiating

(1),

$$\nabla_{\theta} \hat{\ell} = \frac{2}{n^2} \sum_{j,k} v_j v_k S_{jk}, \quad \nabla_{\theta} \ell = \frac{2}{n^2} \sum_j S_{jj} = \frac{2}{n^2} \text{tr } S. \quad (2)$$

The estimator is unbiased for any v with $\mathbb{E}[vv^\top] = I$; its variance is not distribution-free.

Proposition 1 (i.i.d. probes). *Let the components of v be i.i.d. with mean 0, variance 1 and fourth moment κ_4 . Then*

$$\text{Var}(\nabla_{\theta} \hat{\ell}) = \frac{4}{n^4} \left[(\kappa_4 - 1) \sum_j \|S_{jj}\|^2 + \frac{1}{2} \sum_{j \neq k} \|S_{jk} + S_{kj}\|^2 \right]. \quad (3)$$

(Proof in Appendix A; Var of a vector means the trace of its covariance.) Since $\kappa_4 \geq \mathbb{E}[v_i^2]^2 = 1$ with equality iff $|v_i| = 1$ almost surely, Rademacher probes minimise (3) over all i.i.d. laws, and Gaussian probes—the choice in the reference implementation of Rodriguez et al. [2026]—pay exactly $2 \sum_j \|S_{jj}\|^2$ more. We measure this penalty at a factor of 1.18–1.37 depending on the data set (Section 5). It is a free improvement, and we report it separately from everything else so that it is not confused with the contribution of the probe *count*.

Proposition 1 also shows what limits i.i.d. designs. The $\|S_{jk} + S_{kj}\|^2$ term runs over $n^2 - n$ off-diagonal index pairs while the signal $\|\text{tr } S\|^2$ is a sum of only n diagonal terms that need not add coherently; this is why a single Hutchinson probe carries a relative gradient variance of order n , and why no choice of i.i.d. law can do better than a factor 1.23.

Stratified probes. Escaping Proposition 1 requires correlated components. Let $\mathbb{R}^n = \bigoplus_a \mathcal{U}_a$ be an orthogonal decomposition with projectors P_a , dimensions d_a and orthonormal bases Q_a . Fix probabilities $p_a > 0$, $\sum_a p_a = 1$. Draw a stratum $a \sim p$, then

$$v = p_a^{-1/2} Q_a \zeta, \quad \zeta \text{ Rademacher in } \mathbb{R}^{d_a}. \quad (4)$$

Proposition 2 (unbiasedness). *For any orthogonal decomposition and any p with $p_a > 0$, the probe (4) satisfies $\mathbb{E}[vv^\top] = I_n$ exactly, hence $\mathbb{E}[\hat{\ell}] = \ell$ exactly.*

Proof. $\mathbb{E}[vv^\top] = \sum_a p_a \cdot p_a^{-1} Q_a \mathbb{E}[\zeta \zeta^\top] Q_a^\top = \sum_a Q_a Q_a^\top = \sum_a P_a = I$. Substituting into $\mathbb{E}\|\Delta v\|^2 = \text{tr}(\Delta^\top \Delta \mathbb{E}[vv^\top])$ gives $\|\Delta\|_F^2$. $\square$

The decomposition and the probabilities may depend on anything measurable with respect to the past—in particular on previous probes—without disturbing Proposition 2. This is the property that licenses an adaptive design inside an unbiased stochastic gradient method.

Proposition 3 (variance of a stratified design). *With $G_a = Q_a^\top S Q_a$, $T_a = \text{tr}(S P_a)$ and $W_a = \frac{1}{2} \sum_{j \neq k} \|G_{a,jk} + G_{a,kj}\|^2$,*

$$\text{Var}(\nabla_{\theta} \hat{\ell}) = \frac{4}{n^4} \left[\sum_a \frac{W_a + \|T_a\|^2}{p_a} - \left\| \sum_a T_a \right\|^2 \right], \quad (5)$$

minimised over p by $p_a \propto (W_a + \|T_a\|^2)^{1/2}$, giving $\text{Var}^ = \frac{4}{n^4} [(\sum_a \sqrt{W_a + \|T_a\|^2})^2 - \|\sum_a T_a\|^2]$.*

Comparing (5) with (3) at $\kappa_4 = 1$ isolates the trade. Stratification *removes* every cross-stratum term $\|S_{jk} + S_{kj}\|^2$ with j, k in different strata, and *adds* the between-stratum term $\sum_a \|T_a\|^2 / p_a$. Two familiar estimators are the extreme cases: one stratum ($k = 1$) is Hutchinson, and n one-dimensional strata with $p_a = 1/n$ is the one-column estimator $v = \sqrt{n} e_c$, for which every W_a vanishes but the between-stratum term is maximal. That the two ends of this family were found to be statistically indistinguishable in practice [Rodriguez et al., 2026] is consistent with both being off the optimum in opposite directions; the interesting designs are in between, and—crucially—their quality depends on *which* partition is used, not only on how many strata it has.

Alignment, not merely energy. Suppose $\mathcal{U}_1$ is spanned by r leading eigenvectors of the symmetric error Δ . Then $\mathcal{U}_1$ is Δ -invariant, $P_2\Delta P_1 = 0$, and the complement block $G_2 = Q_2^\top \Delta J Q_2 = Q_2^\top P_2 \Delta P_2 J Q_2$ sees only the small tail of the spectrum: W_2 is small, the Neyman rule sends few probes to the complement, and those probes are cheap in variance when they are spent. If instead $\mathcal{U}_1$ merely *captures energy* without being invariant, $P_2\Delta P_1 \neq 0$ leaks the large residual into the complement stratum, W_2 is large, and the gain evaporates. Section 6.1 shows that this distinction is not academic: two sketches built from the same $K = 24$ probes capture the same energy in $r = 16$ directions (93.4% and 92.5% of $\|\Delta\|_F^2$) but leak 3.4% and 5.2% of $\|\Delta P_1\|_F^2$ into the complement, and they differ by $2.5\times$ in the variance they deliver.

A convenient closed form follows if the columns of J are treated as generic vectors of common scale γ , so that $W_a \approx \gamma^2 d_a E_a$ with $E_a = \|\Delta P_a\|_F^2$ the curvature-error energy of stratum a , and the between-stratum term is subdominant. Then $p_a \propto \sqrt{d_a E_a}$ and

$$\frac{\text{Var}_{\text{stratified}}^*}{\text{Var}_{\text{Hutchinson}}} \approx \frac{(\sum_a \sqrt{d_a E_a})^2}{n \|\Delta\|_F^2}. \quad (6)$$

Equation (6) is the design rule we use: it is minimised by making the strata simultaneously low-dimensional and energy-concentrated, which is possible exactly when Δ is low rank.

3 Aligned probing

We now specify the estimator completely. Throughout, B is the minibatch size, $P \leq B$ the probe budget per optimiser step, and all quantities used to build a probe at step t are functions of information available before step t .

3.1 A free sketch of the curvature error

Every probe already computes $\rho = \Delta v = \tilde{H}_\theta v - H v$: it is the vector whose squared norm is the loss. Because $\mathbb{E}[v v^\top] = I$,

$$\mathbb{E}[\rho \rho^\top] = \Delta \mathbb{E}[v v^\top] \Delta^\top = \Delta^2, \quad (7)$$

so the residuals of past probes are a randomised sketch of Δ obtained at zero marginal cost. For each training configuration m we keep a ring buffer of the last $K = 24$ pairs (v_t, ρ_t) , stored in single precision (two ring buffers of $3N \times K$ numbers per configuration, 372 MB in total for the 28,069 training structures).

Let V, R collect the buffered probes and residuals. We split the buffer by push order into an estimation set (three quarters) and a held-out set (one quarter) and use the Nyström factorisation of the estimation set,

$$\Delta \approx R_e (V_e^\top R_e)^+ R_e^\top, \quad (8)$$

whose leading eigenvectors define the stratification basis U . Two properties matter. First, (8) is invariant under $V \mapsto V C$ for any invertible C , so the Horvitz–Thompson rescalings carried by stratified probes do not distort the sketch. Second, and unlike the plainer sketch $\sum_t \rho_t \rho_t^\top$, it uses both V and R and therefore estimates the *eigenvectors* of Δ rather than merely the range of ΔV ; the resulting subspace is an order of magnitude closer to invariant at the same K (Section 6.1).

Diagonalising (8) never touches an $n \times n$ matrix. We form an orthonormal basis Q of $\text{span}(R_e)$ from its $K \times K$ Gram matrix, express the Nyström operator in that basis as $S_c = (Q^\top R_e)(V_e^\top R_e)^+(Q^\top R_e)^\top$, diagonalise the resulting $K \times K$ core, and rotate back. Two batched eigendecompositions of $K \times K$ matrices replace one of an $n \times n$ matrix, a $170\times$ saving that is the difference between a negligible and a dominant overhead (Section 6.3).

Per-direction energies come from the *held-out* residuals, $e_j = \text{mean}_{t \in \text{hold}} (u_j^\top \rho_t)^2$, an unbiased estimate of $\|\Delta u_j\|^2$, and $E_{\text{tot}} = \text{mean}_{t \in \text{hold}} \|\rho_t\|^2$ estimates $\|\Delta\|_F^2$. Cross-fitting is not a refinement: energies read off the same residuals that defined U are biased upwards for exactly the leading directions whose energy sets the stratum probabilities, and using them degrades the design measurably (Section 6.1).

3.2 The probe

For configuration m we use two strata: the leading $r_m = \min(\lceil 0.75 \cdot 3N_m \rceil, \text{rank } R_e, 3N_m - 1)$ directions of U , and their orthogonal complement. With $E_1 = \sum_{j < r_m} e_j$, $E_2 = E_{\text{tot}} - E_1$, $d_1 = r_m$, $d_2 = 3N_m - r_m$, the Neyman rule of (6) gives

$$p_1 = \frac{\sqrt{d_1 E_1}}{\sqrt{d_1 E_1} + \sqrt{d_2 E_2}} \quad \text{clipped to } [0.02, 0.98], \quad (9)$$

and the probe is

$$v = \begin{cases} p_1^{-1/2} U_{:, < r_m} \zeta, & \zeta \text{ Rademacher in } \mathbb{R}^{r_m}, \text{ w.p. } p_1, \\ (1 - p_1)^{-1/2} (z - U_{:, < r_m} U_{:, < r_m}^\top z), & z \text{ Rademacher in } \mathbb{R}^{3N_m}, \text{ w.p. } 1 - p_1. \end{cases} \quad (10)$$

The complement stratum needs no explicit basis. Configurations with fewer than six buffered visits fall back to a plain Rademacher probe. The clip in (9) bounds the inverse-probability weights carried by the probe at 50.

3.3 Which configurations get a probe

Probes are spread over the minibatch by a fixed-size design with inclusion probabilities μ_m and Horvitz–Thompson weights $1/\mu_m$:

$$\hat{\mathcal{L}}_H = \frac{1}{B} \sum_{m \in \mathcal{S}} \frac{1}{\mu_m} \hat{\ell}_m, \quad \sum_m \mu_m = P. \quad (11)$$

The classical choice is Neyman allocation, $\mu_m \propto s_m$, with s_m the per-structure gradient scale; we use the sketch’s free estimate $\hat{\ell}_m = \text{mean}_t \|\rho_t\|^2 / (3N_m)^2$ of the per-structure curvature loss in its place, filtered across visits by a scalar Kalman filter on $\log \ell_m$ whose observation noise is the (known) relative noise of a Hutchinson estimate and whose posterior variance supplies a staleness-driven exploration term. The resulting $\mu_m \propto \hat{s}_m$ are obtained by water filling subject to

$$\mu_{\min} \cdot \frac{P}{B} \leq \mu_m \leq 1, \quad (12)$$

which bounds every Horvitz–Thompson weight by $(B/P)/\mu_{\min}$; the sample is then drawn by Madow systematic π ps sampling from a randomly permuted minibatch, which realises those inclusion probabilities exactly with a deterministic sample size. The floor matters: unbounded weights make the estimator heavy-tailed and cost more variance than the allocation saves (Section 6.1). We report two operating points of (12), both trained: $\mu_{\min} = 1$, which forces $\mu_m \equiv P/B$ and reduces (11) to equal-probability sampling with the constant weight B/P (in that case the score is used only to *order* the structures for systematic sampling, i.e. as implicit stratification), and $\mu_{\min} = 0.25$, which caps the weight at $4B/P$ and is the unequal-probability variant.

Proposition 4 (unbiasedness of the training estimator). *Let $\mathcal{F}_t$ be the σ -algebra generated by everything before step t (so θ_t , the sketches, U , p_1 and the sampling scores are $\mathcal{F}_t$ -measurable). Then $\mathbb{E}[\hat{\mathcal{L}}_H \mid \mathcal{F}_t] = B^{-1} \sum_{m=1}^B \ell_m(\theta_t)$ and $\mathbb{E}[\nabla_\theta \hat{\mathcal{L}}_H \mid \mathcal{F}_t] = \nabla_\theta (B^{-1} \sum_m \ell_m)(\theta_t)$.*

Proof. The inclusion probabilities μ are computed from the filter and sketch states before the sample is drawn, so they are $\mathcal{F}_t$ -measurable, and Madow systematic π ps sampling from a randomly permuted minibatch realises them exactly: the selection indicators $z_m = \mathbf{1}[m \in \mathcal{S}]$ satisfy $\mathbb{E}[z_m \mid \mathcal{F}_t] = \mu_m$. They are drawn independently of the probes, so

$$\mathbb{E}[\hat{\mathcal{L}}_H \mid \mathcal{F}_t] = \frac{1}{B} \sum_{m=1}^B \frac{1}{\mu_m} \mathbb{E}[z_m \mid \mathcal{F}_t] \mathbb{E}[\hat{\ell}_m \mid \mathcal{F}_t] = \frac{1}{B} \sum_{m=1}^B \frac{\mu_m}{\mu_m} \ell_m(\theta_t) = \frac{1}{B} \sum_{m=1}^B \ell_m(\theta_t),$$

where $\mathbb{E}[\hat{\ell}_m \mid \mathcal{F}_t] = \ell_m(\theta_t)$ by Proposition 2, since U and p_1 are $\mathcal{F}_t$ -measurable and ζ, z are fresh. The floor in (12) guarantees $\mu_m > 0$, which is what makes the Horvitz–Thompson weights well defined. The equal-probability operating point is the special case $\mu_m \equiv P/B$. The gradient statement follows because $\mathcal{S}, \mu, U$ and p_1 do not depend on θ , so ∇_θ passes through the finite sum and the two expectations. $\square$

Algorithm 1 Aligned probing: one optimiser step

- 1: **input:** minibatch $\mathcal{B}$ of B structures, probe budget P , per-structure sketch buffers
 - 2: evaluate energies and forces for all B structures {as in E-F training}
 - 3: $\hat{\ell}_m \leftarrow \text{mean}_t \|\rho_t^{(m)}\|^2 / (3N_m)^2$ read off the buffers {free}
 - 4: $\hat{s}_m \leftarrow$ filtered score; $\mu \leftarrow$ water filling of $\mu_m \propto \hat{s}_m$ under (12); $\mathcal{S} \leftarrow$ Madow systematic π ps sample of size P { $\mu_{\min} = 1$ gives $\mu_m \equiv P/B$ and an ordered systematic sample}
 - 5: **for all** $m \in \mathcal{S}$ (**batched**): $U, e \leftarrow$ Nyström subspace (8) and held-out energies; $p_1 \leftarrow$ (9); draw v_m by (10)
 - 6: one batched double-backward returns $\tilde{H}_\theta v_m$ for all $m \in \mathcal{S}$
 - 7: $\rho_m \leftarrow \tilde{H}_\theta v_m - H_m v_m$; $\hat{\mathcal{L}}_H \leftarrow \frac{1}{B} \sum_{m \in \mathcal{S}} \frac{1}{\mu_m} \frac{\|\rho_m\|^2}{(3N_m)^2}$ {Eq. (11)}
 - 8: optimiser step on $\mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \hat{\mathcal{L}}_H$
 - 9: push (v_m, ρ_m) into the buffers of $m \in \mathcal{S}$
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Proposition 4 is unconditional in the sketch: a stale, noisy or outright wrong subspace changes Var but not $\mathbb{E}$. We verified it numerically as well: over 6,000 draws at three budgets the estimator reproduces the exact minibatch curvature loss to within 0.05–0.31%, consistent with Monte-Carlo error, and the audit of Section 5 reports the corresponding check on gradients.

3.4 Achievable probe reduction

Let c be the variance of the aligned estimator at full budget relative to the baseline, and $a = \sum_m \|\nabla \ell_m\|^2 / \text{Var}_{\text{PHL}}$ the share of the baseline variance that is irreducible configuration-to-configuration signal rather than probe noise. Subsampling configurations at rate P/B with constant weights multiplies the probe-noise part by B/P and leaves the signal part, so

$$\frac{\text{Var}_{\text{aligned}}(P)}{\text{Var}_{\text{PHL}}(B)} = \frac{B}{P} (c + a) - a, \quad (13)$$

and a parity requirement $\text{Var}_{\text{aligned}} \leq \tau \text{Var}_{\text{PHL}}$ caps the reduction at $B/P \leq (\tau + a)/(c + a)$. Both c and a are measurable, so (13) is a falsifiable prediction rather than a hope; we test it in Section 6.1. The derivation assumes the constant weights of the $\mu_{\min} = 1$ operating point; unequal probabilities act on the subsampling term itself, so for that rule the same functional form is refitted rather than reused, and we report both fits. It also says where the ceiling comes from: once c is small, the binding term is a , the heterogeneity of the exact per-configuration curvature gradients, which no probe design can touch.

4 Experimental setup

Data and splits. We use the OpenREACT-CHON-EFH data set of ω B97XD/6-31G(d) energies, forces and analytic Hessians [Rodriguez et al., 2025]: 35,087 reactant/transition-state/product (RTP) geometries from 11,954 reactions, 34,248 intrinsic-reaction-coordinate (IRC) geometries, and 62,527 normal-mode-sampled (NMS) geometries; median $N = 14$, maximum $N = 23$. Following Rodriguez et al. [2026], RTP is split 80/10/10 into training, validation and benchmark test sets, and IRC and NMS are held out entirely for evaluation. The split is generated once with a fixed seed and is identical for every run in this paper.

Model, loss and optimiser. We use the authors’ reference implementation [Rodriguez et al., 2026] without modification: a TorchANI [Gao et al., 2020] AEV featuriser with ANI-1x hyperparameters and four per-element $384 \rightarrow 256 \rightarrow 64 \rightarrow 256 \rightarrow 1$ CELU networks; the loss $\mathcal{L}_E + 0.30 \mathcal{L}_F + 0.09 \hat{\mathcal{L}}_H$ with the per-structure normalisations $\mathcal{L}_E \propto (\tilde{E} - E)^2 / \sqrt{N}$, $\mathcal{L}_F \propto \|\tilde{F} - F\|^2 / 3N$ and $\hat{\mathcal{L}}_H \propto \|\Delta v\|^2 / (3N)^2$; AdamW on weights and SGD on biases, both at 10^{-3} , with ReduceLROnPlateau(0.5, patience = 100) on the validation energy RMSE; minibatches of $B = 400$ structures; double precision. All methods share this configuration exactly and differ only in the probe design and the probe count.

Training schedule and its gap to the published regime. Every run in this paper is trained for the same 350 epochs. The plan was 700; running 51 five-seed configurations concurrently on eight GPUs turned out to leave the last 350 epochs outside the available time, so all runs are compared at the common 350-epoch checkpoint that every one of them had already written. Because a checkpoint at epoch 350 does not depend on what happens afterwards, this is exactly a 350-epoch schedule for all methods, and the paired comparisons are unaffected. The published models were trained until the learning-rate schedule terminated, in the vicinity of 700–1000 epochs, and reach lower absolute errors than ours; our reimplemented baseline is therefore under-converged relative to the published numbers, most visibly in energies (Table 1). The gap applies identically to every method here, and Section 6.2 examines whether our conclusions should be expected to survive at full convergence. Every paired comparison in this paper is against our own baseline at the same schedule, data, batch size and seeds; the published numbers appear only as an external anchor.

Protocol, stated in advance. Five seeds (0–4) per configuration, with the seed controlling initialisation, minibatch composition and all probe randomness; the same five seeds for every method. A method *matches* the baseline if a paired t -test over the five seeds gives $p > 0.05$ on *every* one of the nine (dataset $\times$ property) cells; a single rejected cell breaks the match. A method has *variance parity* if the measured gradient-variance ratio is at most 2.0 on each of Test, IRC and NMS. Ratios whose 95% interval straddles 2.0 are reported as failing parity. The headline configuration is the largest probe-count reduction that satisfies both conditions; all budgets we trained are reported regardless of outcome. Probe count is the time-averaged number of probe vectors per optimiser step over the whole run, recorded step by step in the logs.

A noise floor that the seed does not control. Seeds fix initialisation, minibatch composition and probe randomness, but they do not make a run reproducible on this hardware: CUDA reductions in the featuriser are non-deterministic, and over hundreds of SGD steps that amplifies. We measured it, by re-running ten of the reported configurations with identical seeds and code and comparing the last twenty epochs of validation error. The two trajectories agree to machine precision for the first 28–70 epochs and then separate; by epoch 350 they differ by 5.8% on forces, 6.0% on Hessians and 19.6% on energies on average (worst case 32%, 35% and 69%). This is a floor on what any comparison between two runs can resolve here, it is the same order as the detection thresholds quoted in Section 5, and it applies to every method equally. Two consequences are worth stating plainly. First, our paired t -tests are valid but their pairing buys less than it appears to, because the dominant run-to-run noise is not the quantity the pairing controls. Second, a non-rejection is correspondingly easier to obtain, so our accuracy-parity claims should be read as “no difference larger than roughly the run-to-run noise floor” rather than as demonstrated equality. The gradient-variance results are unaffected: they are measured at a fixed checkpoint from many draws of the estimator, not across runs.

Evaluation. Energy and force RMSEs use the complete Test, IRC and NMS sets. Hessian RMSE is the exact full-Hessian RMSE (not a stochastic estimate), computed with the reference implementation’s own `hessian` routine on a fixed subsample of 4,000 structures per evaluation set (the whole Test set is 3,510 structures), identical across methods and seeds.

Gradient-variance audit. At a checkpoint we draw $R = 64$ independent realisations of $\nabla_{\theta} \hat{\mathcal{L}}_H$ per estimator on each of two held-out minibatches of 400 structures and report the trace of the empirical covariance. Because all estimators compared are unbiased for the same minibatch quantity, the ratio of these traces is exactly the ratio of estimator variances and needs no oracle; the exact full-Hessian gradient is computed on one minibatch per audit to verify unbiasedness. On held-out evaluation sets the sketch is warmed by 24 passes before measurement, which is what the estimator’s steady state looks like during training; those warm-up probes are not SGD probes and are logged separately.

5 Results

The reimplementation tracks the published baseline where it matters, and under-converges in energies. At the 350-epoch budget our PHL baseline reaches a Test Hessian RMSE of 15.28 ± 0.19 against 13.38 published, an NMS Hessian RMSE of 36.5 ± 2.7 against 29.71, and Test/IRC/NMS force RMSEs within 10–15% of the published values; energies are the slowest quantity to converge and sit 1.1–1.5 times higher (6.07 ± 0.92 against 4.09 on Test). The energy–force control shows

Table 1: Main results. Probe count is the time-averaged number of probe vectors per optimiser step over the whole run (logged step by step); the reimplemented PHL baseline uses 395.3 (one per molecule, minibatch size $B = 400$ with a short final minibatch). RMSEs are five-seed means, in kcal/mol, kcal/mol/Å and kcal/mol/Å²; standard deviations are in Table 4. p is a paired two-sided t -test against reimplemented PHL over the five shared seeds. The gradient-variance ratio is measured on Test minibatches at the reimplemented-PHL checkpoints (Section 5); per-dataset ratios are in Table 2. ^athe published work reports ensemble means only, so no paired test against it is possible, and it did not run our estimator; ^bthe baseline cannot be tested against itself; ^cenergy–force training sets $\lambda_H = 0$ and therefore has no curvature-loss estimator whose variance could be measured. [†]Figure 3(a) and Figure 4 of Rodriguez et al. [2026], randomised-probe regime; these are cited reference values, not our runs, and were obtained at full convergence on different hardware.

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)	Test E RMSE	Test F RMSE	Test H RMSE	IRC E RMSE	IRC F RMSE	IRC H RMSE	NMS E RMSE	NMS F RMSE	NMS H RMSE	t -test vs. reimpl. PHL (Test E)	Gradient-var ratio (ours / reimpl. PHL)
PHL (paper as reported) [†]	B	13.34	4.09	5.80	13.38	6.91	7.39	15.08	13.54	12.92	29.71	n/a ^a	n/a ^a
Energy–force only (no curvature term)	0.0	2.73	6.92	5.36	151.49	9.21	8.46	151.47	21.75	26.16	136.10	0.46	n/a ^c
One-column probes, B per step	395.3	3.57	7.86	7.23	16.08	10.29	8.66	17.45	17.22	14.71	39.36	0.33	3.05
PHL reimplemented, B per step	395.3	3.81	6.07	6.86	15.28	7.35	8.32	16.72	14.81	14.29	36.49	n/a ^b	1.00
with Rademacher probes (B per step)	395.3	3.72	7.29	7.25	15.99	9.18	8.66	17.29	16.36	14.81	38.89	0.07	0.79
Aligned probing, equal-probability, $P = 160$	164.1	6.69	6.83	7.10	15.61	8.33	8.57	17.04	15.63	14.56	38.36	0.10	1.82
Aligned probing, Neyman allocation, $P = 160$	164.1	3.13	6.91	6.89	15.32	8.54	8.33	16.76	15.55	14.26	37.60	0.53	1.20
Aligned probing, equal-probability, $P = 100$	106.3	6.74	5.72	6.87	15.35	8.61	8.31	16.75	16.00	14.29	37.51	0.52	2.98
Aligned probing, Neyman allocation, $P = 100$	106.3	5.18	6.80	6.91	15.46	10.00	8.36	16.82	17.10	14.30	37.07	0.49	2.06
Aligned probing, equal-probability, $P = 80$	87.1	6.14	5.77	6.94	15.31	8.16	8.35	16.70	15.78	14.33	37.11	0.20	3.77
Aligned probing, equal-probability, $P = 50$	57.9	6.95	5.81	7.14	15.98	8.17	8.55	17.21	15.69	14.54	38.46	0.32	6.37
ablation: no subspace alignment, $P = 100$	106.3	5.85	6.78	7.03	15.69	8.77	8.47	17.01	16.08	14.53	38.21	0.20	4.00
ablation: unordered molecule sample, $P = 100$	106.3	5.39	7.19	7.05	15.57	9.52	8.48	16.91	16.37	14.47	37.90	0.13	3.06

Table 2: Gradient-variance audit. Trace of the empirical covariance of $\nabla_{\theta} \hat{\mathcal{L}}_H$ over $R = 64$ independent draws on held-out minibatches of 400 structures, relative to the reimplemented PHL estimator (Gaussian probes, one per molecule) at the same checkpoint, mean $\pm$ s.d. over the five seeds. The pre-stated parity ceiling is 2.00.

Estimator	probe reduction	Test	IRC	NMS	Train
PHL, Rademacher (B)	$0.99 \times$	0.79 ± 0.02	0.74 ± 0.02	0.73 ± 0.06	0.85 ± 0.04
One-column (B)	$0.99 \times$	3.05 ± 0.23	3.40 ± 0.37	4.41 ± 0.31	2.83 ± 0.16
Aligned, full budget (B)	$0.99 \times$	0.55 ± 0.03	0.49 ± 0.02	0.37 ± 0.03	0.63 ± 0.03
Uniform subsample, isotropic ($P=100$)	$3.95 \times$	4.00 ± 0.30	4.00 ± 0.12	4.18 ± 0.26	4.10 ± 0.19
Aligned, equal-probability ($P=160$)	$2.47 \times$	1.82 ± 0.23	1.67 ± 0.09	1.46 ± 0.24	1.90 ± 0.07
Aligned, Neyman allocation ($P=160$)	$2.47 \times$	1.20 ± 0.16	1.23 ± 0.04	1.10 ± 0.09	1.51 ± 0.16
Aligned, equal-probability ($P=100$)	$3.95 \times$	2.98 ± 0.26	2.99 ± 0.27	2.67 ± 0.18	3.25 ± 0.21
Aligned, Neyman allocation ($P=100$)	$3.95 \times$	2.06 ± 0.37	2.13 ± 0.16	2.06 ± 0.14	2.54 ± 0.22
Aligned, equal-probability ($P=80$)	$4.94 \times$	3.77 ± 0.16	3.88 ± 0.18	3.38 ± 0.32	4.08 ± 0.23
Aligned, equal-probability ($P=50$)	$7.91 \times$	6.37 ± 0.57	6.17 ± 0.48	5.63 ± 0.63	6.70 ± 0.35

the same pattern and reproduces the published qualitative picture: curvature supervision cuts the Hessian RMSE by an order of magnitude ($151.5 \rightarrow 15.3$ on Test, $136.1 \rightarrow 36.5$ on NMS) and roughly halves the NMS force RMSE ($26.2 \rightarrow 14.3$), against published reductions of $139.7 \rightarrow 13.4$ and $24.9 \rightarrow 12.9$. We read our baseline as a faithful but under-converged reproduction, and every statistical claim below is made against it, never against the published numbers.

Probe count. Table 1 reports time-averaged probe counts read from the per-epoch logs. The baseline uses 395.3 probes per optimiser step (one per structure; the last minibatch of each epoch holds 69 structures). Aligned probing at $P = 160, 100, 80, 50$ uses 164.1, 106.3, 87.1 and 57.9, i.e. reductions of $2.41 \times$, $3.72 \times$, $4.54 \times$ and $6.83 \times$. These averages include the eight full-budget warm-up epochs that fill the sketch buffers, which is why they exceed P ; the per-step count is exactly P thereafter, and the warm-up is the spike in Figure 2.

Downstream accuracy. Aligned probing matches the baseline on *all nine* (dataset $\times$ property) cells at $P = 160$ with either allocation rule, and at $P = 100$ and $P = 80$ with equal probabilities: no paired t -test over the five shared seeds rejects at $p \leq 0.05$ (Table 3). The six force and Hessian cells are tightly matched—at the headline setting the largest deviation on any of them is $+3.1\%$, and the five-seed standard deviation of those cells is $0.6\text{--}7.3\%$ of their mean, so the test has real power there (the smallest difference it could resolve is 4.0% on IRC force and Hessian, $5.5\text{--}5.6\%$ on Test

Table 3: Paired t -tests against reimplemented PHL on every (dataset $\times$ property) cell, over the five shared seeds. Each entry is the mean relative RMSE difference in per cent (positive = worse than PHL) with the two-sided p -value in parentheses; * marks $p \leq 0.05$.

Method	Test			IRC			NMS		
	E	F	H	E	F	H	E	F	H
Neyman allocation $P=160$	+13.9(0.53)	+0.5(0.82)	+0.3(0.90)	+16.2(0.22)	+0.1(0.94)	+0.2(0.87)	+5.0(0.28)	-0.2(0.91)	+3.0(0.45)
Equal-probability $P=160$	+12.5(0.10)	+3.5(0.27)	+2.1(0.40)	+13.3(0.17)	+3.0(0.26)	+1.9(0.36)	+5.6(0.20)	+1.9(0.23)	+5.1(0.08)
Neyman allocation $P=100$	+12.0(0.49)	+0.8(0.27)	+1.2(0.27)	+36.0(0.06)	+0.5(0.33)	+0.6(0.24)	+15.5(0.04)*	+0.1(0.94)	+1.6(0.54)
Equal-probability $P=100$	-5.8(0.52)	+0.3(0.78)	+0.4(0.47)	+17.1(0.15)	-0.1(0.95)	+0.2(0.80)	+8.0(0.10)	+0.0(1.00)	+2.8(0.19)
Equal-probability $P=80$	-4.9(0.20)	+1.2(0.12)	+0.1(0.84)	+11.0(0.21)	+0.4(0.52)	-0.1(0.89)	+6.6(0.15)	+0.3(0.74)	+1.7(0.53)
Equal-probability $P=50$	-4.4(0.32)	+4.2(0.00)*	+4.6(0.04)*	+11.1(0.07)	+2.8(0.01)*	+2.9(0.12)	+6.0(0.14)	+1.8(0.37)	+5.4(0.35)
No alignment $P=100$	+11.8(0.20)	+2.5(0.33)	+2.7(0.23)	+19.3(0.17)	+1.8(0.36)	+1.7(0.35)	+8.6(0.12)	+1.7(0.16)	+4.7(0.00)*
Unordered sample $P=100$	+18.5(0.13)	+2.8(0.07)	+1.9(0.00)*	+29.5(0.01)*	+1.9(0.13)	+1.2(0.09)	+10.6(0.01)*	+1.2(0.53)	+3.9(0.30)
PHL + Rademacher (B)	+20.1(0.07)	+5.8(0.32)	+4.6(0.38)	+24.9(0.20)	+4.1(0.36)	+3.4(0.41)	+10.5(0.30)	+3.7(0.44)	+6.6(0.43)
Energy-force only	+14.0(0.46)	-21.8(0.00)*	+891.2(0.00)*	+25.3(0.29)	+1.7(0.29)	+805.9(0.00)*	+46.9(0.00)*	+83.1(0.00)*	+273.0(0.00)*

Table 4: RMSEs of Table 1 with standard deviations over seeds. Units: kcal/mol, kcal/mol/Å, kcal/mol/Å².

Method	Test			IRC			NMS		
	E	F	H	E	F	H	E	F	H
Energy-force only ($n=5$)	6.92 $\pm$ 2.13	5.36 $\pm$ 0.24	151.49 $\pm$ 2.51	9.21 $\pm$ 3.12	8.46 $\pm$ 0.24	151.47 $\pm$ 2.39	21.75 $\pm$ 1.14	26.16 $\pm$ 0.08	136.10 $\pm$ 1.33
One-column probes (B) ($n=5$)	7.86 $\pm$ 3.28	7.23 $\pm$ 0.32	16.08 $\pm$ 0.73	10.29 $\pm$ 4.30	8.66 $\pm$ 0.31	17.45 $\pm$ 0.71	17.22 $\pm$ 3.52	14.71 $\pm$ 0.53	39.36 $\pm$ 3.09
PHL reimplemented (B) ($n=5$)	6.07 $\pm$ 0.92	6.86 $\pm$ 0.04	15.28 $\pm$ 0.19	7.35 $\pm$ 0.64	8.32 $\pm$ 0.06	16.72 $\pm$ 0.12	14.81 $\pm$ 0.80	14.29 $\pm$ 0.35	36.49 $\pm$ 2.67
PHL + Rademacher (B) ($n=5$)	7.29 $\pm$ 1.60	7.25 $\pm$ 0.80	15.99 $\pm$ 1.68	9.18 $\pm$ 2.33	8.66 $\pm$ 0.74	17.29 $\pm$ 1.42	16.36 $\pm$ 2.19	14.81 $\pm$ 1.44	38.89 $\pm$ 6.29
Equal-probability $P=160$ ($n=5$)	6.83 $\pm$ 0.97	7.10 $\pm$ 0.39	15.61 $\pm$ 0.62	8.33 $\pm$ 1.83	8.57 $\pm$ 0.42	17.04 $\pm$ 0.60	15.63 $\pm$ 1.90	14.56 $\pm$ 0.66	38.36 $\pm$ 2.09
Aligned probing $P=100$ ($n=5$)	5.72 $\pm$ 0.61	6.87 $\pm$ 0.12	15.35 $\pm$ 0.16	8.61 $\pm$ 1.14	8.31 $\pm$ 0.13	16.75 $\pm$ 0.18	16.00 $\pm$ 0.90	14.29 $\pm$ 0.12	37.51 $\pm$ 1.45
Aligned probing $P=80$ ($n=5$)	5.77 $\pm$ 0.56	6.94 $\pm$ 0.08	15.31 $\pm$ 0.18	8.16 $\pm$ 0.90	8.35 $\pm$ 0.06	16.70 $\pm$ 0.16	15.78 $\pm$ 1.26	14.33 $\pm$ 0.26	37.11 $\pm$ 1.61
Aligned probing $P=50$ ($n=5$)	5.81 $\pm$ 0.74	7.14 $\pm$ 0.10	15.98 $\pm$ 0.61	8.17 $\pm$ 0.66	8.55 $\pm$ 0.10	17.21 $\pm$ 0.60	15.69 $\pm$ 1.06	14.54 $\pm$ 0.26	38.46 $\pm$ 1.87
Ablation: no alignment $P=100$ ($n=5$)	6.78 $\pm$ 0.50	7.03 $\pm$ 0.31	15.69 $\pm$ 0.47	8.77 $\pm$ 2.10	8.47 $\pm$ 0.29	17.01 $\pm$ 0.48	16.08 $\pm$ 1.69	14.53 $\pm$ 0.46	38.21 $\pm$ 2.89
Ablation: unordered sample $P=100$ ($n=5$)	7.19 $\pm$ 2.04	7.05 $\pm$ 0.16	15.57 $\pm$ 0.19	9.52 $\pm$ 1.26	8.48 $\pm$ 0.14	16.91 $\pm$ 0.20	16.37 $\pm$ 1.13	14.47 $\pm$ 0.27	37.90 $\pm$ 0.73

force and Hessian, and 4.5 and 10.2% on NMS force and Hessian). The three energy cells are not: their seed-to-seed standard deviation is 5–15% of the mean, the smallest resolvable difference is 11% (NMS), 31% (IRC) and 56% (Test), and the observed differences—+5.0%, +16.2% and +13.9% at the headline setting—are well inside that band. We report those three cells as not rejected, which is what the pre-stated criterion asks, but a non-rejection on a cell whose detection threshold is 56% is weak evidence of equivalence and we do not present it as strong. Energies are the slowest quantity to converge in this architecture and the one our shortened schedule hurts most; the force and Hessian cells, which are what curvature supervision is for, carry the weight of the accuracy claim. Three configurations do break the match. At $P = 50$ (6.83 $\times$), Test force (+4.2%, $p = 0.002$), IRC force (+2.8%, $p = 0.008$) and Test Hessian (+4.6%, $p = 0.038$) are resolved. The unequal-probability rule at $P = 100$ is resolved on one cell, NMS energy (+15.5%, $p = 0.040$), which is the noisiest cell in the study (a five-seed s.d. of 1.8 kcal/mol on a mean of 17) but which we score as a failure under the all-cell rule we stated. And the ablation that draws the molecule sample without ordering it by the free score is resolved on three cells, which is what promotes score-ordered systematic sampling from a refinement to a component. For reference the energy-force control differs on six of nine cells and the one-column estimator on one (NMS Hessian, +7.9%, $p = 0.029$); switching the baseline’s probes from Gaussian to Rademacher changes nothing measurable, as Proposition 1 requires—it changes the variance, not the estimand.

Gradient variance. Table 2 reports the audit, and three things in it should be separated. First, at *equal* probe count the aligned design reduces the variance of the curvature-loss gradient to 0.55 ± 0.03 (Test), 0.49 ± 0.02 (IRC) and 0.37 ± 0.03 (NMS) of the baseline, i.e. by 1.8 to 2.7 times, of which the free switch from Gaussian to Rademacher accounts for 0.79, 0.74 and 0.73. Second, the one-column estimator is 3.1 to 4.4 times *worse* than Gaussian Hutchinson on the gradient although its downstream accuracy is nearly indistinguishable—a dissociation we return to. Third, at reduced budget the ratio grows, and how fast depends on the allocation rule. With equal probabilities it is 1.82 ± 0.23 (Test), 1.67 ± 0.09 (IRC) and 1.46 ± 0.24 (NMS) at $P = 160$, and 2.98, 2.99, 2.67 at $P = 100$; with unequal probabilities it is 1.20 ± 0.16 , 1.23 ± 0.04 and 1.10 ± 0.09 at $P = 160$ and 2.06, 2.13, 2.06 at $P = 100$. Applying the convention of Section 4, the 95% intervals of the five-seed mean at $P = 160$ are [1.01, 1.40], [1.18, 1.28] and [0.98, 1.21] for the unequal-probability rule—all three clear the ceiling of 2.0 with margin—against [1.53, 2.11], [1.56, 1.78] and [1.16, 1.76] for equal probabilities, where Test straddles the ceiling and is scored as not clearing. At $P = 100$ the

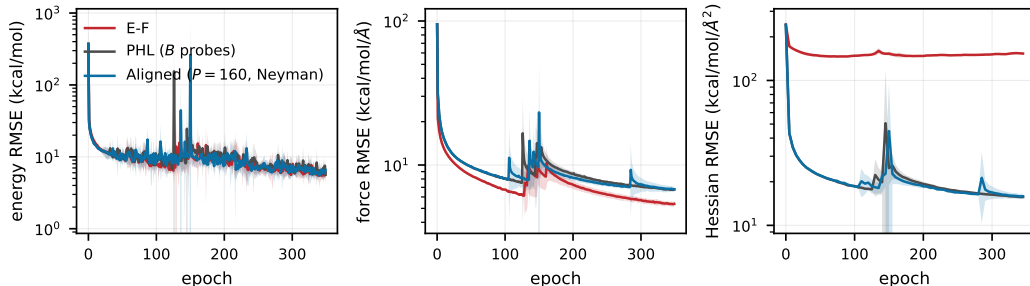


Figure 1: Validation RMSE against epoch, mean $\pm$ s.d. over five seeds. Aligned probing at the headline setting ($P = 160$ with unequal-probability allocation, a $2.41\times$ probe-count reduction) tracks the B -probe baseline on energies, forces and the exact Hessian throughout training; the energy–force control never recovers curvature.

unequal-probability rule sits on the boundary ($[1.60, 2.52]$, $[1.93, 2.33]$, $[1.88, 2.23]$) and is scored as not clearing; every budget below that fails on all three sets. Empirical unbiasedness is confirmed at the same time: the mean of the 64 draws differs from the exact full-Hessian gradient by 0.9 to 1.3 standard errors for every estimator, including ours—exactly the spread expected if all of them are unbiased.

Headline. The largest budget satisfying both pre-stated conditions is $P = 160$ with unequal-probability allocation: a $2.41\times$ time-averaged probe-count reduction, accuracy indistinguishable from the reimplemented baseline on all nine cells, and gradient variance of 1.20, 1.23 and 1.10 times the baseline’s on Test, IRC and NMS, every interval well inside the ceiling. The equal-probability rule reaches the same probe count and the same accuracy but only 1.82, 1.67, 1.46 in variance, with Test straddling. This is a modest rather than a large reduction, and Section 3.4 says why: the ceiling $(\tau + a)/(c + a)$ with the constants fitted to each rule evaluates at $\tau = 2$ to $2.74\times$, $2.82\times$ and $3.13\times$ for equal probabilities and $3.85\times$, $3.74\times$ and $3.87\times$ for unequal ones. The next budget we trained, $3.72\times$, is just past both predictions, and misses on variance for both rules—as predicted.

An informative dissociation. Accuracy parity survives to $4.54\times$ while per-step variance parity is lost beyond $2.41\times$ (the $3.72\times$ point is within 6% of the ceiling and still matches accuracy on all nine cells under the equal-probability rule). As a proxy for what training needs, the per-step criterion is therefore conservative: Adam averages the curvature-gradient noise over the hundreds of steps between learning-rate reductions, and a fourfold increase in a noise term that enters the total loss with weight $\lambda_H = 0.09$ is largely filtered out. We report both because they answer different questions—the variance ratio characterises the estimator, the t -tests characterise the trained model—and because the criterion we set ourselves in Section 4 was the stricter one. Figure 1 shows the corresponding validation curves: at $P = 100$ the run tracks the baseline on all three targets throughout training.

6 Analysis

6.1 Mechanism

The curvature error is low rank, and more so out of distribution. Diagonalising Δ exactly for 200 structures per data set at the baseline’s 350-epoch checkpoint gives mean stable ranks $\|\Delta\|_F^2/\|\Delta\|_2^2$ of 3.76 (train), 3.79 (Test), 3.57 (IRC) and 3.22 (NMS) against $3N \approx 41$, with the leading fifth of the directions carrying about three quarters of $\|\Delta\|_F^2$ (Figure 4, left). The ordering is the one the mechanism predicts: NMS, the far-from-equilibrium set on which the model is worst, is also the set on which its error is most concentrated, and it is where aligned probing delivers its largest variance reduction (0.37 against 0.55 on Test at equal probe count). Chemically this is what one expects—large-amplitude normal-mode displacements excite a few stiff, strongly anharmonic directions that a smooth atom-centred potential systematically mis-curves—and it means that a

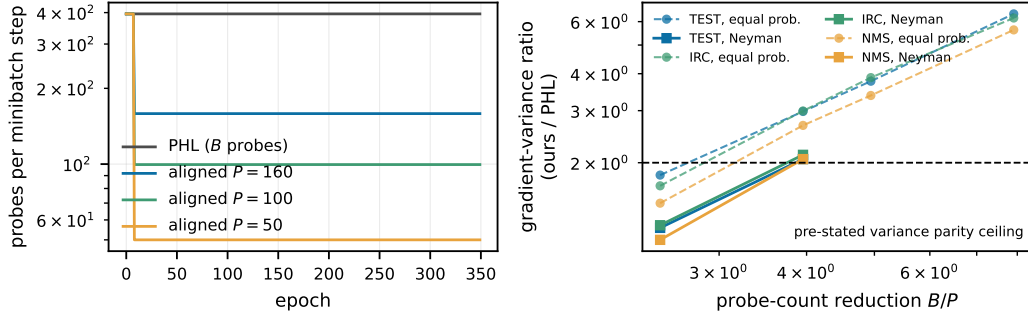


Figure 2: Left: probes per optimiser step over training (seed 0); the spike at the left edge is the eight full-budget warm-up epochs that fill the sketch buffers. Right: measured gradient-variance ratio against probe-count reduction, mean over five seeds, for the equal-probability rule (dashed) and the unequal-probability rule (solid); the horizontal dashed line is the pre-stated parity ceiling. The abscissa is the per-step ratio B/P ; the time-averaged reduction reported in Table 1 is 2–3% smaller because of the warm-up epochs.

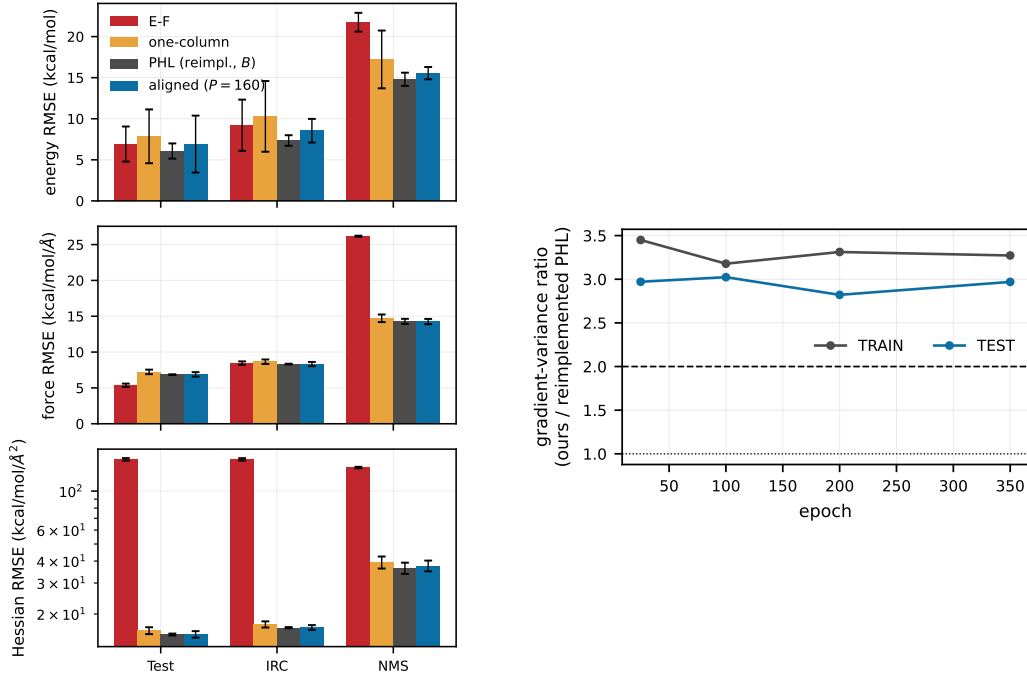


Figure 3: Left: final RMSEs on the three evaluation sets, mean $\pm$ s.d. over five seeds. Right: gradient-variance ratio of aligned probing at $P = 100$ (equal-probability rule) against training epoch, measured at the baseline’s own checkpoints; the alignment gain is flat over a fourteen-fold range of training progress.

method which spends probes where the error is helps most exactly where curvature supervision is most needed.

Near-invariance, not captured energy, is what the design needs. Figure 4 (right) compares four candidate stratification bases at equal cost. The eigenbasis of the *reference* Hessian H —free, exact and never stale—captures only 42% of $\|\Delta\|_F^2$ in eight directions against 85% for the exact eigenbasis of Δ : the model’s curvature error is not concentrated in the stiffest physical modes, a negative result worth recording because that basis would otherwise be the obvious choice. The two sketch-based

bases are almost indistinguishable by captured energy (93.4% against 92.5% at $r = 16$) but differ by a factor 1.5–1.9 in leakage $\|P_2\Delta P_1\|_F^2/\|\Delta P_1\|_F^2$ (0.034 against 0.052 at $r = 16$, 0.060 against 0.116 at $r = 8$), and in a controlled measurement at fixed checkpoint and fixed K they differ by $2.5\times$ in delivered gradient variance (0.27 against 0.68). This is the prediction of Section 2 made concrete: what the complement stratum inherits is $P_2\Delta P_2$, and only a near-invariant subspace keeps that small.

The sketch does not degrade by feeding on its own probes. A natural worry about a self-supervised design is a feedback loop: probes are drawn preferentially inside the current subspace, the sketch therefore sees the complement rarely, and its estimate of the complement decays. Re-running the audit with the sketch filled by *isotropic* probes instead changes nothing measurable—0.602 against 0.609 (train), 0.517 against 0.536 (Test), 0.371 against 0.398 (NMS) at full budget—so the Horvitz–Thompson rescaling that keeps the estimator unbiased also keeps the sketch unbiased, and the 20–30% of probes that the Neyman rule sends to the complement suffice to maintain it.

Which component carries the result. At $P = 100$, removing the subspace alignment entirely (uniform molecule subsampling with isotropic Rademacher probes) costs a factor 1.34 in variance on Test and IRC and 1.56 on NMS (4.00, 4.00 and 4.18 against 2.98, 2.99 and 2.67), and its downstream accuracy is resolved away from the baseline on NMS Hessian (+4.7%, $p = 0.0007$). Ordering the equal-probability molecule sample by the free per-structure score moves the estimator variance by less than 3% in either direction (3.06 against 2.98 on Test, 2.96 against 2.99 on IRC, 2.67 against 2.67 on NMS)—yet the five-seed trained ablation without the ordering is resolved on three accuracy cells against none with it. Systematic sampling on an ordered list is not merely a variance device: it also removes the run-to-run variability in *which* structures are probed, and the accuracy tests are evidently sensitive to that even where the per-step variance is not. The ranking of the three components by effect on the estimator is alignment (1.8–2.7 $\times$), unequal-probability allocation (1.3–1.5 $\times$), ordering ($< 1.03\times$); by effect on trained accuracy, alignment and ordering both matter and allocation does not.

Allocating probes across molecules: a smaller effect than alignment, and one we initially measured with the wrong sign. The textbook answer to a probe budget is Neyman allocation, $\mu_m \propto s_m$, de-biased with weights $1/\mu_m$; the achievable gain is $1/(1 + \text{CV}^2)$ in the coefficient of variation of the per-structure gradient scale across a minibatch. Our first measurement of that quantity, on a pilot checkpoint 10 epochs into training, gave $\text{CV} = 0.51$, a ceiling of only $1.26\times$; with the inclusion probabilities left unbounded the Horvitz–Thompson weights reached 50, the estimator became heavy-tailed, and allocation lost outright (at $P = 200$, 5.22 against 1.86 for equal probabilities). We concluded that the molecule axis was not worth allocating along, and built the method around the direction axis.

That conclusion was half wrong, and the audit at the trained checkpoints says why. Heterogeneity is not a constant of the problem: by epoch 350 the coefficient of variation of s_m has grown to 0.74 (Test), 0.76 (train) and 0.84 (NMS), raising the ceiling to 1.55, 1.57 and 1.71 $\times$; and the free score $\hat{\ell}_m$ tracks the true per-structure gradient scale with Spearman correlation 0.77–0.81, so most of that ceiling is reachable. Once the inclusion probability is floored at $\mu_{\min} = 0.25$ of the uniform value—bounding the weight at $4B/P$ —the allocation delivers 1.52 $\times$, 1.36 $\times$ and 1.33 $\times$ on Test, IRC and NMS at $P = 160$ and 1.45 $\times$, 1.40 $\times$, 1.30 $\times$ at $P = 100$, inside the ceiling in every case. What was actually wrong was not the idea but the unbounded weights, and a pilot measurement taken too early in training.

The comparison with the direction axis nevertheless stands, and remains the point of the paper. Allocation across molecules is worth about 1.3–1.5 $\times$; alignment across directions is worth 1.8–2.7 $\times$ at equal probe count and is the difference between matching and not matching downstream accuracy at $P = 100$. Per-structure gradient scales are moderately heterogeneous ($\text{CV} \approx 0.8$); per-direction curvature errors inside a single structure are far more so (stable rank 3.8 out of 41). Both axes should be used; the second is where the leverage is.

The variance model predicts the ceiling. Fitting Eq. (13) to the four equal-probability budgets on each data set gives $(c, a) = (0.55, 0.29)$ on Test, $(0.49, 0.35)$ on IRC and $(0.37, 0.39)$ on NMS, hence predicted maximum reductions at $\tau = 2$ of 2.74 $\times$, 2.82 $\times$ and 3.13 $\times$. Unequal-probability allocation acts on the subsampling term itself, so both constants must be refitted; its two budgets give $(0.36, 0.22)$, $(0.34, 0.27)$ and $(0.14, 0.50)$ and ceilings of 3.85 $\times$, 3.74 $\times$ and 3.87 $\times$. Both predictions are borne out: the largest budget we trained that satisfies both conditions is 2.41 $\times$, and 3.72 $\times$ —just

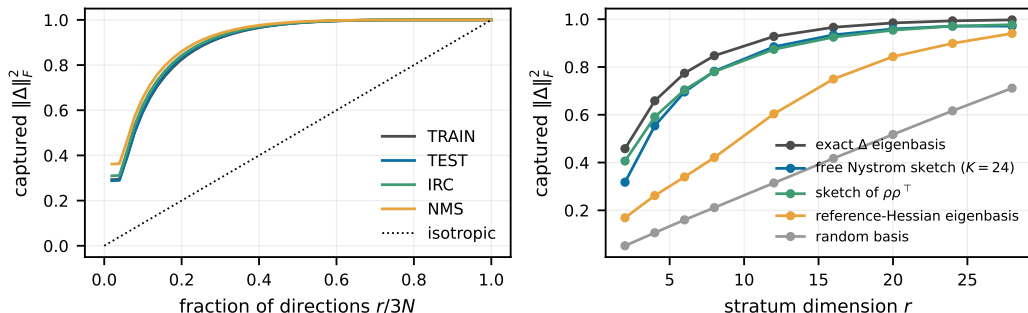


Figure 4: Left: cumulative fraction of $\|\Delta\|_F^2$ carried by the leading exact eigendirections of the curvature error, averaged over 200 structures per data set at the baseline’s 350-epoch checkpoint; the dotted line is what an isotropic error would give. Right: fraction captured by the leading r directions of each candidate stratification basis, on the training set.

below the unequal-probability ceiling and well above the equal-probability one—misses on variance for both rules while still matching accuracy for one of them. The model also says what would have to change to do better. Driving c to zero would move the ceiling only to $\tau/a \approx 4\text{--}9\times$: beyond that the binding term is a , the dispersion of the *exact* per-structure curvature gradients across a minibatch, a property of the data and the loss that no probe design touches. A larger reduction under this criterion therefore requires either a larger minibatch or a criterion stated on the accumulated rather than the per-step gradient.

6.2 Where the method fails

Cells and budgets where the method loses. At $P = 50$ ($6.83\times$) three cells are resolved away from the baseline—Test force +4.2%, IRC force +2.8%, Test Hessian +4.6%—and the accuracy condition fails. The pattern is interpretable: those are the quantities on which the baseline is most accurate in absolute terms and which therefore have the least room to absorb extra gradient noise, whereas NMS, where errors are three to seven times larger, holds up best at every budget in variance. Accuracy on NMS is a different story: it is also where the two configurations that fail on a single cell fail (NMS energy for unequal-probability allocation at $P = 100$, NMS Hessian for the no-alignment ablation), because NMS energy is by some margin the noisiest cell in the study—its five-seed standard deviation is 1.8 kcal/mol against a between-method spread of about 2. We score those failures as failures, but a reader should treat single-cell NMS-energy rejections with more caution than the coherent three-cell pattern at $P = 50$ —and, symmetrically, should not read the non-rejections on Test and IRC energy as demonstrated equivalence, since at five seeds those cells could not resolve a difference below 31 and 56 per cent (Section 5). The honest summary is that this study has the statistical power to establish force and Hessian parity to within roughly the fixed-seed noise floor of 6% (Section 4) and only to bound energy parity loosely. Closing that gap needs more seeds and a deterministic featuriser, not more probes: with a per-run noise floor of 6% on forces and Hessians, resolving a 2% effect would take of order a hundred runs per configuration. The variance condition fails from $P = 100$ onwards on all three sets and for both allocation rules.

The Test set is the hardest of the three, and the reason is structural. At every budget the variance ratio is ordered $\text{NMS} < \text{IRC} < \text{Test} \lesssim \text{Train}$, and that ordering follows the stable rank of the curvature error ($3.22 < 3.57 < 3.79 \approx 3.76$). Test consists of optimised reactants, transition states and products—structures at or near stationary points, where the model has been trained hardest and where its residual error is the least structured of the three sets. A design that buys variance reduction from anisotropy necessarily buys least where the error is closest to isotropic. This is a statement about which chemistry the idea suits, not a defect of the implementation: expect the largest gains on strained, reactive and far-from-equilibrium geometries and the smallest on relaxed ones.

Would the gains transfer to full convergence? Our baseline is under-converged in energies relative to the published models (Section 4), so the honest question is whether the mechanism is convergence-stage-dependent. The two components answer differently, and we have measurements for both. The *alignment* gain is flat: the equal-budget variance ratio measured at epochs 25, 100, 200 and 350 of the baseline moves only between 0.57 and 0.61 on the training set and between 0.54 and 0.57 on Test (Figure 3, right), and the stable rank of Δ —the structural quantity it depends on—is flat as well, so we expect it to transfer. The *allocation* gain is not flat, and this is the clearest convergence-stage dependence we found: the coefficient of variation of the per-structure gradient scale grows from 0.51 at epoch 10 to 0.74–0.84 at epoch 350, raising the ceiling on the allocation gain from $1.26\times$ to $1.55\text{--}1.71\times$. A model trained to convergence is one on which most structures are fit well and a few are not, which is exactly the regime that rewards unequal allocation; if anything we expect this component to be *stronger* at full convergence, not weaker. What we cannot verify is the remaining term: at full convergence the exact per-structure gradients shrink, which changes a in Eq. (13) and hence the achievable reduction, in a direction our data cannot resolve. A run to learning-rate termination would settle it and is the first experiment we would add.

A limitation we could not remove in time. An earlier prototype of the same design, which diagonalises the full $3N \times 3N$ Nyström operator rather than its $K \times K$ core and realises the complement stratum in an explicit orthonormal basis rather than as $(I - P_U)z$, reaches an equal-budget variance ratio of 0.33 against the 0.63 of the implementation we trained, at $40\times$ the per-step cost. We identified the gap late. Proposition 1 explains part of it—the projector realisation has near-Gaussian coefficients in any basis of the complement and pays the $(\kappa_4 - 1)$ term—but substituting an in-basis complement into the fast implementation recovered only part of the difference, and we did not resolve the remainder before the budget for retraining ran out. Everything reported here is the fast implementation: it is what was trained and what was audited, and the prototype is recorded in the attempts log. If the gap is real and closable, Eq. (13) with $c = 0.33$ would move the ceiling from $2.7\times$ to about $4\times$ on Test.

6.3 Computational cost

Aligned probing is designed to leave the expensive part of the step untouched, and to add only arithmetic that is cheap relative to a double-backward. The measurements say it succeeds on both counts, and also that at this system size the probe saving does not turn into a wall-clock saving. It is worth being precise about all three.

Where the time goes. On an idle H200 with $B = 400$ in double precision, one optimiser step costs 10.7 ms for the energy–force graph alone and 23.3 ms with the full-budget curvature term. Restricting the double-backward to P probed structures by building a second autograd graph costs 31.4 ms and is *independent of P* between $P = 20$ and $P = 200$: the workload is bound by kernel launches rather than arithmetic, so a double-backward over 20 structures costs what one over 400 costs, and the second model invocation is pure overhead. We therefore train with a single graph in which the probe vector is exactly zero on unprobed structures. That is mathematically identical, enters exactly P probe vectors into the estimator, and costs the same per step as the baseline.

The cost of the design itself. Building the Nyström subspace, the cross-fitted energies and the stratum probabilities costs 1.9–2.3 ms per step, essentially independent of P , against 92 ms if the $3N \times 3N$ operator were diagonalised instead—the one optimisation that makes the design viable. Molecule selection, systematic sampling on the free per-structure score, costs 0.21 ms. Measured end to end, on one otherwise idle GPU, an epoch takes 0.864 s for energy–force training, 1.747 s for the baseline, 1.759 s for probe subsampling without alignment, 1.885 s at $P = 100$ and 1.952 s at $P = 160$. Aligned probing at the headline budget therefore costs 12% more wall-clock than the baseline while using $2.4\times$ fewer probes: the classical overhead of the allocation machinery is real but small, which is the regime the introduction argued for. (The per-epoch times in Table 1 come from the training logs, where up to fifty runs shared eight GPUs; they are inflated by contention and are not comparable across methods. The controlled numbers here are the ones to read, and are logged separately.)

What this means for the metric. The probe saving does not become a wall-clock saving here, and we do not claim one. Probe count and wall-clock are different quantities at this scale: a probe

is an operation whose cost grows with system size, whereas the launch overhead that dominates our timings is a property of 14-atom molecules on a current GPU. Our sub-batched implementation, whose double-backward cost genuinely scales with P , becomes the faster of the two once a double-backward over B structures stops being launch-bound—at this model size, several thousand structures per minibatch. Finally, the one change that *would* buy wall-clock in a launch-bound regime—skipping the curvature term on whole steps rather than on structures within a step—is strictly worse statistically at equal time-averaged probe count, because it concentrates all the sampling noise in time.

7 Related work

Curvature supervision for interatomic potentials. Training on second derivatives is an instance of Sobolev training [Czarnecki et al., 2017] and has recently become practical for interatomic potentials through Hessian–vector products. Rodriguez et al. [2026] formulate the Hutchinson curvature loss used here and compare Gaussian probes with the single-column probe $\sqrt{3Ne_c}$ used by Amin et al. [2025] and Cui et al. [2026]; Koker et al. [2026] apply stochastic curvature supervision to phonon fine-tuning in periodic materials, and Yuan et al. [2024] and Williams et al. [2025] study analytic Hessians from learned potentials and the reference data needed to fit them. All of this work draws probes isotropically. Our contribution is orthogonal to the choice of architecture or loss: it concerns the probe distribution, and it applies unchanged to any of these schemes.

Stochastic trace estimation. The variance of Hutchinson’s estimator and the advantage of Rademacher over Gaussian probes are classical [Hutchinson, 1990, Avron and Toledo, 2011, Roosta-Khorasani and Ascher, 2015], and Proposition 1 is the gradient analogue of those results. Modern estimators such as Hutch++ [Meyer et al., 2021], its variants [Persson et al., 2022] and XTrace [Epperly et al., 2024] improve on Hutchinson by deflating a low-rank sketch, computed from $O(r)$ *additional* matrix–vector products, and then probing the remainder. Our design differs in three respects that the setting forces. First, the target is not a trace but the *gradient* of a trace, and a deflation term built from a stale sketch is constant in θ and therefore contributes nothing to the gradient; deflation in the Hutch++ sense cannot help here, whereas stratification can. Second, the sketch is not purchased: it is assembled from residuals the loss has already produced, and is amortised over the epochs in which a structure is revisited, so the estimator has the same per-step cost as plain Hutchinson. Third, the operator is not fixed: Δ moves as the model trains, which makes an unbiased-for-any-sketch design (Proposition 2) necessary rather than merely convenient. The Nyström factorisation we use for the sketch is standard [Williams and Seeger, 2001, Gittens and Mahoney, 2016, Halko et al., 2011]; what is new is the observation that in this setting it is free, and that it is the right sketch because near-invariance, not captured energy, is what the stratified variance depends on.

Adaptive sampling in stochastic optimisation. Allocating effort across training examples by importance sampling has a long literature [Needell et al., 2014, Zhao and Zhang, 2015, Alain et al., 2015, Katharopoulos and Fleuret, 2018, Johnson and Guestrin, 2018, Csiba and Richtárik, 2018], as do control-variate schemes such as SVRG and SAGA [Johnson and Zhang, 2013, Defazio et al., 2014]. The obstacle in that literature is that per-example gradient norms are expensive, so surrogates are used. Our setting hands us an exact, free surrogate—the residual norm $\|\rho_m\|^2$ —and we report in Section 6.1 that even with it, the gain from unequal-probability allocation is modest (1.3–1.5 $\times$), is negative unless the Horvitz–Thompson weights are explicitly bounded, and grows during training as the model’s errors concentrate on fewer structures. The allocation dimension that pays most here is not which example to probe but which *direction* to probe, a degree of freedom that has no counterpart in first-order supervised learning and appears only because the supervised object is an operator.

8 Conclusion

Stochastic curvature supervision has so far been treated as a question of how many probes one can afford. We have argued that it is at least as much a question of where to put them. The curvature error of a trained interatomic potential is not isotropic: its stable rank is under four in a space of forty-one dimensions, and it is most concentrated precisely on the far-from-equilibrium geometries where curvature supervision matters most. An isotropic Hutchinson probe spends most of its variance budget on directions in which the model is already right.

The estimator that follows is cheap and safe. It is exactly unbiased for any stratification and any stratum probabilities, so the subspace it aligns to may be stale, noisy or wrong without ever biasing the gradient, and the information needed to align it is a by-product the loss has already computed, so the alignment is free. At equal probe count it reduces the variance of the curvature-loss gradient by a factor of 1.8 to 2.7; trading that surplus for probes, and combining it with a bounded unequal-probability allocation across structures worth a further 1.3 to 1.5, gives a $2.4\times$ reduction in probes per optimiser step at a gradient variance of 1.10 to 1.23 times the baseline’s and accuracy indistinguishable from it on all nine evaluated cells. Accuracy parity in fact survives to $4.5\times$, past the point where the per-step variance criterion is lost.

We have been equally explicit about the limits. A variance model with two measurable constants predicts the ceiling we hit and identifies what sets it: past a certain point the binding term is not probe noise but the dispersion of the exact per-structure curvature gradients, which no probe design can reduce. Allocating the budget across molecules rather than across directions is worth roughly a fifth as much, and only once the inverse-probability weights it introduces are bounded—a conclusion our own early measurements got backwards, and which we correct here rather than quietly drop. And the reduction we obtain is modest rather than transformative: $2.4\times$, not the order of magnitude the low-rank structure might suggest at first sight.

The most promising direction is not a better subspace estimator but a better criterion. Per-step variance parity is a conservative proxy, as our own results show: accuracy survives twice as far. A criterion stated on the gradient accumulated over the window an optimiser actually averages—the quantity that determines the trajectory—would license substantially larger reductions from the same design and would put the accounting on what training is really sensitive to.

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A Proofs

Proposition 1. Write $g = \nabla_{\theta} \hat{\ell}$ and drop the factor $2/n^2$. From (2), $g = \sum_{jk} v_j v_k S_{jk}$ and $\mathbb{E}g = \sum_j S_{jj}$. For i.i.d. components with $\mathbb{E}v_i = 0$, $\mathbb{E}v_i^2 = 1$, $\mathbb{E}v_i^4 = \kappa_4$, the fourth moment $\mathbb{E}[v_j v_k v_l v_m]$ is κ_4 when all four indices coincide, 1 when the indices form two distinct equal pairs (the three cases $j = k \neq l = m$, $j = l \neq k = m$, $j = m \neq k = l$), and 0 otherwise. Hence

$$\mathbb{E}\|g\|^2 = \kappa_4 \sum_j \|S_{jj}\|^2 + \sum_{j \neq l} \langle S_{jj}, S_{ll} \rangle + \sum_{j \neq k} \|S_{jk}\|^2 + \sum_{j \neq k} \langle S_{jk}, S_{kj} \rangle.$$

Since $\|\mathbb{E}g\|^2 = \sum_j \|S_{jj}\|^2 + \sum_{j \neq l} \langle S_{jj}, S_{ll} \rangle$, subtracting gives $\text{Var}(g) = (\kappa_4 - 1) \sum_j \|S_{jj}\|^2 + \sum_{j \neq k} (\|S_{jk}\|^2 + \langle S_{jk}, S_{kj} \rangle)$. Finally, summing $\|S_{jk} + S_{kj}\|^2 = \|S_{jk}\|^2 + \|S_{kj}\|^2 + 2\langle S_{jk}, S_{kj} \rangle$ over ordered pairs $j \neq k$ shows that $\sum_{j \neq k} (\|S_{jk}\|^2 + \langle S_{jk}, S_{kj} \rangle) = \frac{1}{2} \sum_{j \neq k} \|S_{jk} + S_{kj}\|^2$, which is manifestly non-negative. Restoring $2/n^2$ gives (3). Because $\kappa_4 - 1 = \text{Var}(v_i^2) \geq 0$ vanishes only when v_i^2 is almost surely constant, and $\mathbb{E}v_i^2 = 1$, the minimum over unit-variance i.i.d. laws is attained exactly at $|v_i| = 1$ a.s., i.e. at the Rademacher law. $\square$

Proposition 3. Condition on the stratum a . Then $v = p_a^{-1/2} Q_a \zeta$ and $\sum_{jk} v_j v_k S_{jk} = p_a^{-1} \zeta^\top G_a \zeta$ with $G_a = Q_a^\top S Q_a$, so by Proposition 1 applied inside stratum a (where ζ is Rademacher, $\kappa_4 = 1$), $\mathbb{E}[g | a] = p_a^{-1} \text{tr } G_a = p_a^{-1} T_a$ and $\text{Var}(g | a) = p_a^{-2} W_a$ with $W_a = \frac{1}{2} \sum_{j \neq k} \|G_{a,jk} + G_{a,kj}\|^2$; note $\text{tr } G_a = \text{tr}(Q_a Q_a^\top S) = \text{tr}(S P_a)$. The law of total variance gives $\mathbb{E}\|g\|^2 = \sum_a p_a (p_a^{-2} W_a + p_a^{-2} \|T_a\|^2) = \sum_a (W_a + \|T_a\|^2)/p_a$, while $\mathbb{E}g = \sum_a T_a$, which is (5). Minimising $\sum_a \alpha_a/p_a$ subject to $\sum_a p_a = 1$ with $\alpha_a = W_a + \|T_a\|^2$ gives $p_a \propto \sqrt{\alpha_a}$ and optimum $(\sum_a \sqrt{\alpha_a})^2$ by Cauchy–Schwarz. $\square$

Remark (both realisations of a stratum probe). Proposition 2 holds equally for $v = p_a^{-1/2} P_a z$ with z Rademacher in $\mathbb{R}^n$, since $\mathbb{E}[P_a z z^\top P_a] = P_a$. We use the basis form $Q_a \zeta$ for the leading stratum, where the basis is available, and the projector form for the complement, where it saves constructing a basis; the two differ only in fourth moments, and by Proposition 1 the basis form is the better of the two when a basis is at hand.